Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/01/2012
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marshall.long@yale.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-432-4229
Organization Name:	Yale University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	PO Box 208284		
PI Web Page:			
City:	New Haven	**State:**	CT
Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2006	**End Date:**	12/31/2011
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS ; Flight definition NOTE: Extended to 12/31/2011, per NSSC info (Ed., 9/27/2011) NOTE: Extended to 9/30/2011, per NSSC info (9/2009) NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell (Yale University)		
Grant/Contract No.:	NNX07AB33A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNC04AA03A, Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment, with grants reinstatement in 2006. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Computational modeling The computational models used in our early flame studies included the chemical mechanisms of both GRI Mech 2.11 and a simple 26-species, C2 hydrocarbon mechanism developed at Yale. Both mechanisms clearly showed that buoyancy plays a role in both the size and shape of the laminar diffusion flame. Good comparisons between computations and experiments were obtained using the 26-species C2 mechanism for our standard, moderately diluted flame consisting of 65% methane and 35% nitrogen. However, the agreement in lift-off heights deteriorated rapidly as the dilution level in the flame was increased. Analysis of this disagreement pointed to a need for improved chemical kinetics. A 35-species chemical kinetic mechanism was used to study formaldehyde concentrations in our standard flame configuration under normal gravity conditions with an eye toward understanding the relationship of formaldehyde production and heat release rates. A recent computational effort re-analyzed our standard flames at both normal and zero gravity using a more up-to-date kinetics scheme based on one of the more comprehensive and well-validated sets available for ethylene. The resultant mechanism contains 476 reactions and 66 chemical species. Another area that requires further study in a microgravity environment is sooty flames. Despite the complexities of modeling soot formation in flames using detailed chemistry, the linkage of soot production to radiation and bulk flame properties is so strong that the coupled treatment of this problem has become a necessity for quantitative modeling of flame structure. Refinements to an earlier version of our soot model have recently been implemented. In this more recent work, as in a number of flame studies, the soot volume fraction was predicted using a sectional representation. The model assumes that soot particles can be approximated as carbon spheres and that they exist in the free molecular limit (i.e., the mean free path of the gaseous mixture is much larger than the largest soot spheroid). The approach specifies a minimum and maximum particle mass and divides the spheroid sizes logarithmically into a number of bins or sections. A further assumption is that within a given section, spheroids of varying diameter do not exhibit vastly differing aerosol dynamical qualities. This assumption holds as long as the number of sections is sufficiently large. In our most recent computations, a total of 30 soot sections were employed (rather than 20 in previous studies). The surface growth model used in the numerical simulations was also modified. Quantitative Measurements with Simple Diagnostics Our recent experimental efforts have concentrated on further refinement and characterization of diagnostics and flame configurations that will allow us to take greatest advantage of the long periods of high-quality microgravity that are available on the International Space Station (ISS). We have worked to develop techniques for making the comparison between experiments and computations as quantitative as possible, even with relatively simple diagnostic techniques. A technique utilizing a color digital camera has been developed that allows observation of flame shape, size, lift-off height, and measurements of temperature via soot and thin-filament pyrometry as well as quantitative determination of the soot volume fraction. Techniques for obtaining an absolute light calibration have also been developed, which is essential for measuring the absolute soot volume fraction and the number density of chemiluminescent species such as CH*. Microgravity Experiments: SLICE and CLD Flame Our Coflow Laminar Diffusion Flame experiment (CLD Flame) is one of the four investigations that make up ACME. The experiments in ACME are currently in development for conduct within the Combustion Integrated Rack (CIR) on the ISS. As a precursor to CLD Flame, the Structure & Liftoff in Combustion Experiment (SLICE) takes advantage of the existence of a closely related set of experiments recently completed on board the ISS. The goal of SLICE is to investigate the structure of lifted and sooty flames. The results can provide a test case for developing the computational model and soot sub-model, and also be used to minimize risk and maximize the scientific return of the CLD Flame Experiment. Another milestone for the ACME experiments was reached in May, 2010 when the Requirements Definition Review (RDR) was held in Cleveland, Ohio. The review process included a Science Review Board and a Technical Review Board. The Technical Review Board assessed “the state of the ACME requirements and engineering development as being largely at the state of readiness needed to proceed to preliminary design.” In discussing the CLD Flame experiment, the RDR Science Panel report stated “the experiments will provide a unique database for testing combustion

	simulations. Long time periods at low gravity are needed to reach steady-state flame conditions. These conditions allow for more precise measurements and simplify the comparison of simulation and measurement results. The camera-based, imaging diagnostics developed at Yale have been thoroughly tested and will provide a unique, high-quality data set for use in this project and in future simulation projects.”
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Dworkin SB, Cooke JA, Bennett BAV, Connelly BC, Long MB, Smooke MD, Hall RJ, Colket MB. "Distributed-memory parallel computation of a forced, time-dependent, sooting, ethylene/air coflow diffusion flame. " Combustion Theory and Modelling. 2009;13(5):795-822. http://dx.doi.org/10.1080/13647830903159293 , Nov-2009
Articles in Peer-reviewed Journals	Herdman JD, Connelly BC, Smooke MD, Long MB, Miller JH. "A comparison of Raman signatures and laser-induced incandescence with direct numerical simulation of soot growth in non-premixed ethylene/air flames." Carbon. 2011 Dec;49(15):5298-311. http://dx.doi.org/10.1016/j.carbon.2011.07.050 , Dec-2011
Articles in Peer-reviewed Journals	Kuhn PB, Ma B, Connelly BC, Smooke MD, Long MB. "Soot and thin-filament pyrometry using a color digital camera." Proceedings of the Combustion Institute 2011;33(1):743-50. http://dx.doi.org/10.1016/j.proci.2010.05.006 , Jan-2011
Dissertations and Theses	Connelly BC. "Quantitative Characterization of Steady and Time-Varying, Sooting, Laminar Diffusion Flames using Optical Techniques." Ph.D. Thesis, Yale University, 2009. , May-2009
Papers from Meeting Proceedings	Ma B, Cao S, Bennett BAV, Smooke MD, Long MB. "Experimental and computational study of lifted coflow laminar diffusion flames under elevated pressures." 7th US National Technical Meeting of the Combustion Institute, Atlanta, GA, March 20-23, 2011. In: 7th US National Technical Meeting of the Combustion Institute, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1652-1657. Printed by Curran Associates, Inc. ISBN: 978-1-61782-780-8. , Jun-2011
Papers from Meeting Proceedings	Cao S, Bennett BAV, Ma B, Long MB, Smooke MD. "Computational and experimental study of laminar coflow methane-air diffusion flames under elevated pressures." 2011 Fall Technical Meeting of the Eastern States Section of the Combustion Institute, Storrs, CT, October 9-12, 2011. 2011 Fall Technical Meeting of the Eastern States Section of the Combustion Institute, Storrs, CT, October 9-12, 2011. B-21. , Oct-2011
Papers from Meeting Proceedings	Ma B, Long MB. "Absolute light calibration in combustion experiments." 2011 Fall Technical Meeting of the Eastern States Section of the Combustion Institute, Storrs, CT, October 9-12, 2011. 2011 Fall Technical Meeting of the Eastern States Section of the Combustion Institute, Storrs, CT, October 9-12, 2011. B-34. , Oct-2011